

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q16-0650

Mount Storm 138 kV
August 30, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0650. The Interconnection Customer proposes a 1500.0 MW natural gas combined cycle facility interconnecting at the Mount Storm 138 kV substation in the West Virginia area. The request was received on July 19, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 307 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$218,574,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Mount Storm to Santee Cooper 138 kV	Mount Storm to Jamaica 138 kV line	662.7	226	100.0
Jamaica to Mount Storm 138 kV	Mount Storm to Santee Cooper 138 kV	634.1	236	100.0
Cypress to Windom 138 kV Line	Mount Storm to Santee Cooper 138 kV	474.1	190	59.8
Windom to Morbihan 138 kV L	Mount Storm to Santee Cooper 138 kV	418.2	211	59.8
Cypress Substation to Coleman Road 138 kV Line	Mount Storm to Santee Cooper 138 kV	411.6	224	59.8
Coleman Road to Planta Santurce 138 kV	Mount Storm to Santee Cooper 138 kV	390.5	240	59.8
Camphora to Jamaica 138 kV	Santee Cooper to Mount Storm 138 kV Line	390.1	241	59.8
Planta Santurce to Camphora 138 kV	Santee Cooper to Mount Storm 138 kV line	389.7	241	59.8
Jamaica Substation to Digton 138 kV Line	Mount Storm to Santee Cooper 138 kV Line	276.1	200	40.2
Deep Creek to Dighton 138 kV Line	Mount Storm to Santee Cooper 138 kV Line	270.1	203	40.2
New Creek Wind to Midulla Generating Station 138 kV L	Santee Cooper 345-138 kV Autotransformer	269.0	305	53.1
Midulla Generating Sta to Bel Cottage Substation 138 kV	Santee Cooper Substation 345 kV/138 kV Transformer Bank	263.5	310	53.1
New Creek Wind to Brandon Mres 138 kV	Santee Cooper 345 kV/138 kV Autotransformer	263.5	313	53.1

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Brandon Mres to Santee Cooper 138 kV	Santee Cooper 345/138 kV Transformer Bank	240.2	343	53.1
Bell Cottage to Kahe 2 138 kV L	Santee Cooper Substation 345/138 kV Transformer Bank	220.5	189	32.3
Decker Substation to Deep Crk 138 kV	Mount Storm to Santee Cooper 138 kV	215.8	211	33.3
Santee Cooper 345/138 kV Transformer	Brandon Mres to Santee Cooper 138 kV L	195.7	547	76.2
Glen Head to N Shore Beach 138 kV	Santee Cooper 345-138 kV Auto	152.7	248	20.9
Bell Cottage to North Shore Beach Substation 138 kV	Santee Cooper 345/138 kV Autotransformer	145.0	266	20.9
Metcalf Energy Center to Kahe 2 138 kV	Greenland Gap to Santee Cooper 345 kV	139.4	345	11.1
Decker Substation to Edenville Substation 138 kV	Santee Cooper Substation to Mount Storm 138 kV L	133.1	212	22.5
St. John to Rio Dell 138 kV Line	Amercan Substation to Urban 345 kV	120.9	369	7.3
Edenville to Green Bluff 138 kV	Santee Cooper to Mount Storm 138 kV Line	120.0	221	22.5
Greenland Gap to Santee Cooper 345 kV	Jamaica to Mount Storm 138 kV	109.0	1,700	44.6
N E to Deker 138 kV	Mount Storm to Santee Cooper 138 kV	107.8	237	17.6

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Mount Storm to Santee Cooper 138 kV	Rebuild line with higher-capacity conductor	\$2,224,000	100% (\$2,224,000)
Munt Storm to Jamaica 138 kV	Rebuild line with higher-capacity conductor	\$24,000,000	100% (\$24,000,000)
Windom to Cypress 138 kV	Rebuild line with higher-capacity conductor	\$3,600,000	100% (\$3,600,000)
Windom to Morbihan 138 kV	Rebuild line with higher-capacity conductor	\$3,721,000	100% (\$3,721,000)
Cypress Substation to Coleman Road 138 kV Line	Rebuild line with higher-capacity conductor	\$2,556,000	100% (\$2,556,000)
Planta Santurce to Coleman Road 138 kV	Rebuild line with higher-capacity conductor	\$3,059,000	100% (\$3,059,000)
Jamaica Substation to Camphora 138 kV L	Rebuild line with higher-capacity conductor	\$6,502,000	100% (\$6,502,000)
Camphora to Planta Santurce 138 kV	Rebuild line with higher-capacity conductor	\$6,336,000	100% (\$6,336,000)
Jamaica Substation to Digton 138 kV Line	Rebuild line with higher-capacity conductor	\$8,001,000	100% (\$8,001,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Dighton to Deep Crk 138 kV Line	Rebuild line with higher-capacity conductor	\$12,431,000	100% (\$12,431,000)
Midulla Generating Station to New Crk Wind 138 kV	Rebuild line with higher-capacity conductor	\$10,131,000	100% (\$10,131,000)
Midulla Generating Sta to Bel Cottage Substation 138 kV	Rebuild line with higher-capacity conductor	\$5,940,000	100% (\$5,940,000)
Brandon Mres to New Crk Wind 138 kV	Rebuild line with higher-capacity conductor	\$4,725,000	100% (\$4,725,000)
Brandon Mres to Santee Cooper 138 kV	Rebuild line with higher-capacity conductor	\$2,915,000	100% (\$2,915,000)
Bell Cottage to Kahe 2 138 kV L	Rebuild line with higher-capacity conductor	\$5,747,000	100% (\$5,747,000)
Deep Crk to Decker 138 kV L	Rebuild line with higher-capacity conductor	\$16,373,000	100% (\$16,373,000)
Santee Cooper 345/138kV XFMR	Replace transformer with higher-rated unit	\$13,866,000	100% (\$13,866,000)
Glen Head to N Shore Beach 138 kV	Rebuild line with higher-capacity conductor	\$8,117,000	100% (\$8,117,000)
Bell Cottage to North Shore Beach Substation 138 kV	Rebuild line with higher-capacity conductor	\$9,782,000	100% (\$9,782,000)
Metcalf Energy Center to Kahe 2 138 kV	Rebuild line with higher-capacity conductor	\$28,861,000	93% (\$26,732,000)
Decker to Edenville 138 kV	Rebuild line with higher-capacity conductor	\$19,218,000	94% (\$18,072,000)
St. John to Rio Dell 138 kV	Rebuild line with higher-capacity conductor	\$11,836,000	100% (\$11,836,000)
Edenville to Green Bluff 138 kV	Rebuild line with higher-capacity conductor	\$7,813,000	94% (\$7,347,000)
Greenland Gap to Santee Cooper 345 kV	Reconductor line	\$3,223,000	71% (\$2,302,000)
N E to Decker 138 kV	Reconductor line	\$2,259,000	100% (\$2,259,000)
	Total allocated to project		\$218,574,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.